

VISUAL EXPLORATION OF HEART DISEASE IN THE CLEVELAND DATASET - Homework 4

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How do cardiovascular stress indicators interact with demographic factors to predict heart disease presence?

```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyr)
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'
```

```
## The following object is masked from 'package:dplyr':
##
##   combine
```

```
library(scales)
```

Load the Cleveland Heart Disease dataset

```
# Load data from URL
url <- "https://archive.ics.uci.edu/ml/machine-learning-databases/heart-disease/processed.cleveland.data"
```

```
# Column names based on UCI documentation
col_names <- c("age", "sex", "cp", "trestbps", "chol", "fbs",
               "restecg", "thalach", "exang", "oldpeak",
               "slope", "ca", "thal", "target")

# Load directly from the URL
heart_data <- read.csv(url, header = FALSE, col.names = col_names, na.strings = "?")

cat("Data loaded successfully from URL\n")
```

```
## Data loaded successfully from URL
```

```
# Data preprocessing
heart_data <- heart_data %>%
  mutate(
    # Convert target to binary: 0 = no disease, 1-4 = disease present
    disease = ifelse(target > 0, "Disease", "Healthy"),
    disease = factor(disease, levels = c("Healthy", "Disease")),

    # Convert categorical variables to factors with meaningful labels
    sex = factor(sex, levels = c(0, 1), labels = c("Female", "Male")),

    cp_type = factor(cp, levels = c(1, 2, 3, 4),
                     labels = c("Typical Angina", "Atypical Angina",
                               "Non-anginal", "Asymptomatic")),

    fbs = factor(fbs, levels = c(0, 1),
                 labels = c("FBS ≤ 120", "FBS > 120")),

    exang = factor(exang, levels = c(0, 1),
                  labels = c("No Ex. Angina", "Ex. Angina")),

    restecg_type = factor(restecg, levels = c(0, 1, 2),
                          labels = c("Normal", "ST-T Abnormality", "LV Hypertrophy")),

    slope_type = factor(slope, levels = c(1, 2, 3),
                        labels = c("Upsloping", "Flat", "Downsloping")),

    # Age groups for better visualization
    age_group = cut(age, breaks = c(0, 40, 50, 60, 70, 100),
                    labels = c("<40", "40-50", "50-60", "60-70", "70+"))
  ) %>%
  filter(!is.na(ca) & !is.na(thal)) # Remove missing values

# Define custom color palette
disease_colors <- c("Healthy" = "#2ecc71", "Disease" = "#e74c3c")
sex_colors <- c("Female" = "#9b59b6", "Male" = "#3498db")
```

Task: Relationship between Sex and Heart Disease

```
# Convert target to binary: 0 = no disease, 1 = disease
heart_data$target <- ifelse(heart_data$target > 0, 1, 0)

# Convert sex to factor for modeling
heart_data$sex <- factor(heart_data$sex, labels = c("female", "male"))

# Run a simple logistic regression model
model <- glm(target ~ sex, data = heart_data, family = "binomial")

# Display summary
summary(model)
```

```
##
## Call:
## glm(formula = target ~ sex, family = "binomial", data = heart_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.0438      0.2326  -4.488 7.18e-06 ***
## sexmale       1.2737      0.2725   4.674 2.95e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 409.95  on 296  degrees of freedom
## Residual deviance: 386.12  on 295  degrees of freedom
## AIC: 390.12
##
## Number of Fisher Scoring iterations: 4
```

Figure 1 - Age and Maximum Heart Rate Relationship by Disease Status

Figure 1 shows this through the lens of maximum heart rate—how fast your heart can beat during intense exercise. But here's where it gets interesting: people with heart disease don't just slow down a little; they slow down dramatically compared to healthy people their same age.

Look at the red points (disease patients) versus the green ones (healthy patients). At every age, diseased hearts simply can't keep up. A healthy 50-year-old might hit 160-170 beats per minute on a stress test, while someone with heart disease might only reach 130-140. That's not just a small difference—it's a window into how well the heart is actually functioning.

The gap between the two groups gets wider as people age? This tells us something important: heart disease doesn't just subtract a fixed amount from your capacity. It changes how your heart ages. The triangles represent males and circles represent females, and you can see this pattern affects both sexes, though our dataset has more men than women.

FIGURE 1: Age and Maximum Heart Rate Relationship by Disease Status

fundamental relationship between age, cardiovascular capacity, and disease presence with sex as an additional dimension

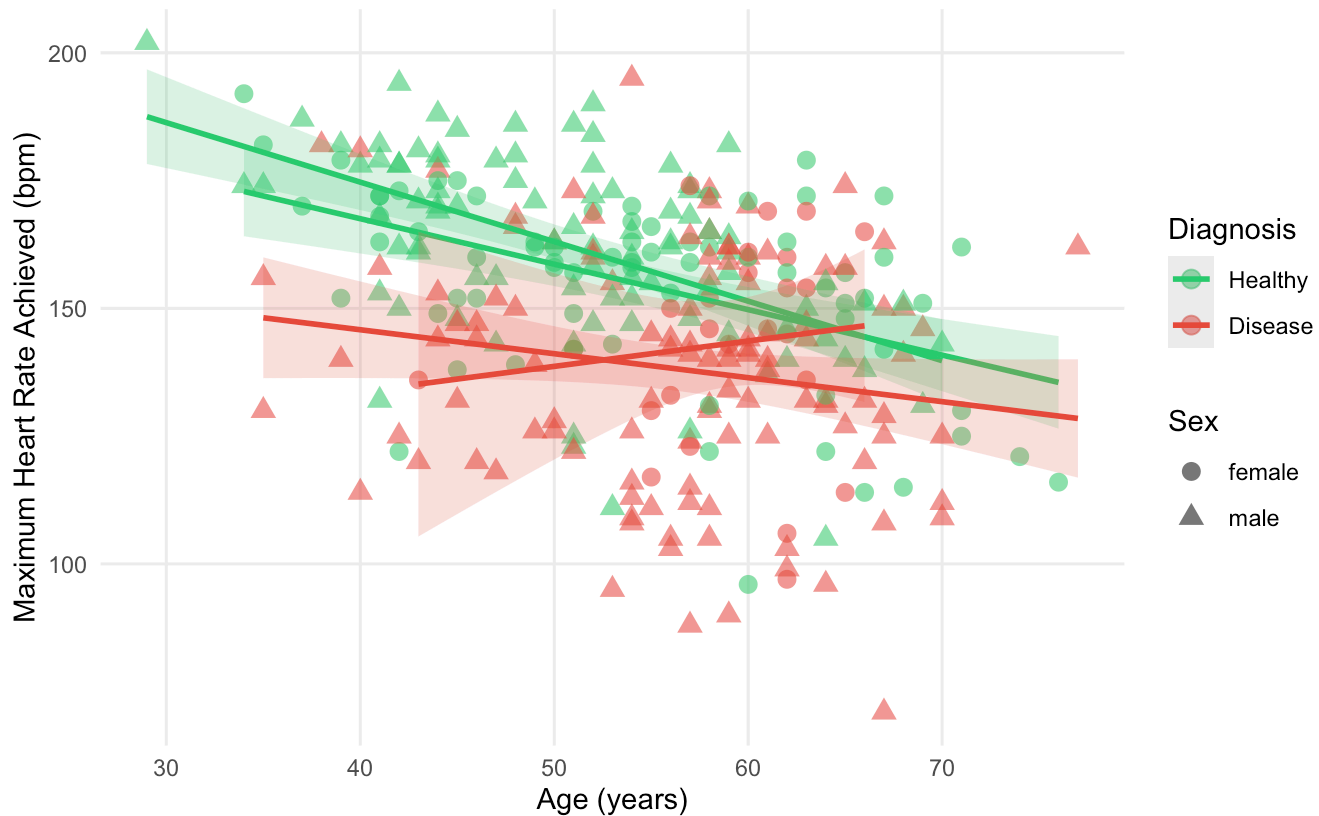
```
fig1 <- ggplot(heart_data, aes(x = age, y = thalach, color = disease, shape = sex)) +
  geom_point(alpha = 0.6, size = 3) +
  geom_smooth(method = "lm", se = TRUE, aes(fill = disease), alpha = 0.2) +
  scale_color_manual(values = disease_colors, name = "Diagnosis") +
  scale_fill_manual(values = disease_colors, guide = "none") +
  scale_shape_manual(values = c(16, 17), name = "Sex") +
  labs(
    title = "Age vs. Maximum Heart Rate: The Cardiovascular Capacity Decline",
    subtitle = "Heart disease patients show reduced maximum heart rate across all ages",
    x = "Age (years)",
    y = "Maximum Heart Rate Achieved (bpm)",
    caption = "Each point represents one patient; lines show linear trends with 95% CI"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 10, color = "gray40"),
    legend.position = "right",
    panel.grid.minor = element_blank()
  )

print(fig1)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Age vs. Maximum Heart Rate: The Cardiovascular Capacity Decline

Heart disease patients show reduced maximum heart rate across all ages



Each point represents one patient; lines show linear trends with 95% CI

```
ggsave("figure1_age_hearttrate.png", fig1, width = 10, height = 6, dpi = 300)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Figure 2 - Chest Pain Type Distribution Across Sex and Disease Status

The patients with the *worst* heart disease often don't feel chest pain at all. But Figure 2 reveals a disturbing paradox, especially for men.

Look at the right panel showing males. The tall red bars under "Asymptomatic"? That's the highest concentration of heart disease patients—and they reported *no chest pain*. This is why doctors can't just ask "Does your chest hurt?" and call it a day.

For women (left panel), the pattern is a bit different, with symptoms more evenly spread across different pain types. But people experiencing "typical angina"—that classic crushing chest pain everyone warns you about—are often actually *healthy*. Their pain might be from anxiety, acid reflux, or muscle strain, not their heart.

This figure says **symptoms (or lack thereof) might be lying to you**. This is why doctors insist on objective tests. Some of the sickest hearts beat in the chests of people who feel perfectly fine, while some perfectly healthy hearts belong to people convinced they're having a heart attack.

If there's one figure that explains why we can't rely on "how you feel" for heart disease diagnosis, this is it.

```
# FIGURE 2: Chest Pain Type Distribution Across Sex and Disease Status
```

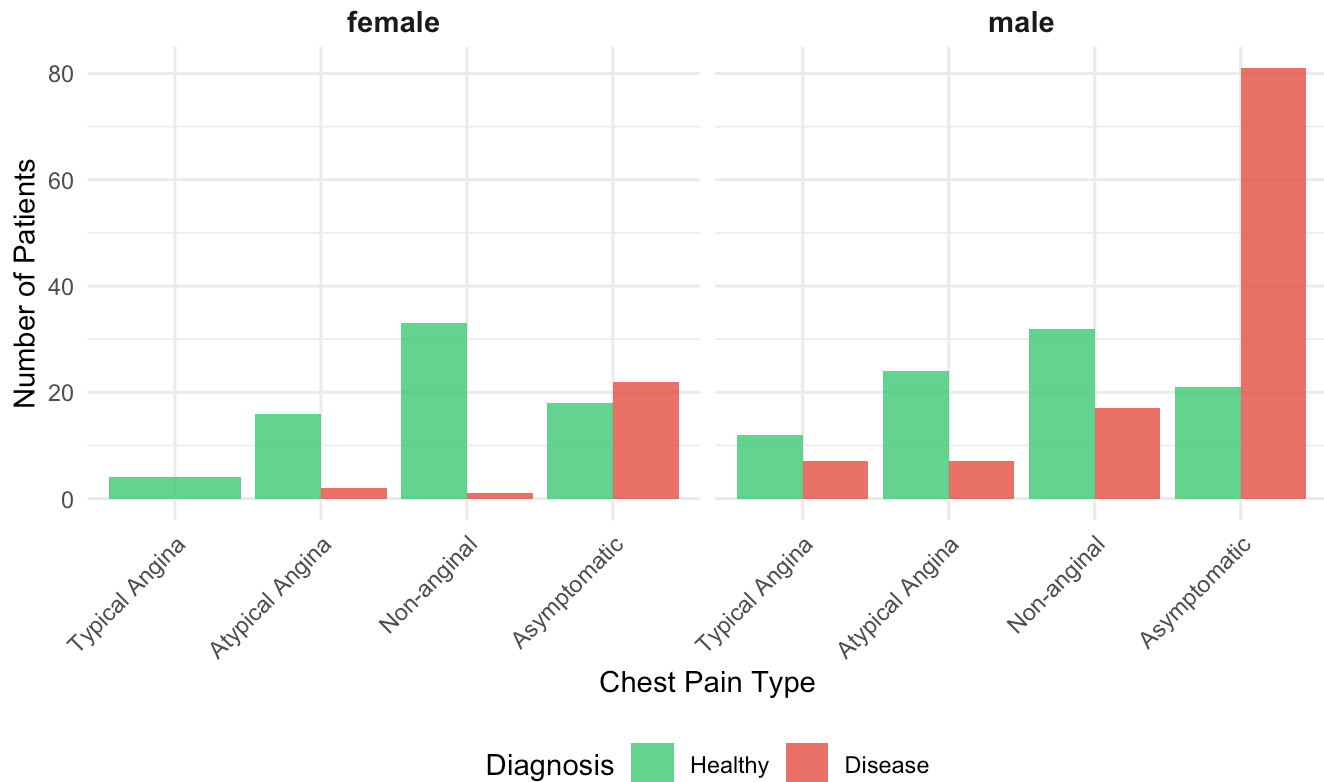
```
# Reveals critical patterns in symptom presentation
```

```
fig2 <- heart_data %>%
  group_by(cp_type, sex, disease) %>%
  summarise(count = n(), .groups = "drop") %>%
  ggplot(aes(x = cp_type, y = count, fill = disease)) +
  geom_bar(stat = "identity", position = "dodge", alpha = 0.8) +
  facet_wrap(~sex, ncol = 2) +
  scale_fill_manual(values = disease_colors, name = "Diagnosis") +
  labs(
    title = "Chest Pain Patterns: A Critical Diagnostic Paradox",
    subtitle = "Asymptomatic patients show highest disease prevalence, especially in males",
    x = "Chest Pain Type",
    y = "Number of Patients",
    caption = "Note: Asymptomatic presentation is a major risk factor"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 10, color = "gray40"),
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.text = element_text(face = "bold", size = 11),
    legend.position = "bottom"
  )

print(fig2)
```

Chest Pain Patterns: A Critical Diagnostic Paradox

Asymptomatic patients show highest disease prevalence, especially in males



Note: Asymptomatic presentation is a major risk factor

```
ggsave("figure2_chestpain_distribution.png", fig2, width = 10, height = 6, dpi = 300)
```

Figure 3 - ST Depression and Blood Pressure Interaction

Figure 3 examines what happens when we stress the heart through exercise, measuring ST depression (an ECG marker of inadequate blood flow) against resting blood pressure.

The left panel shows patients without exercise-induced chest pain—here, healthy and diseased individuals overlap considerably. But the right panel tells a different story: patients who experience chest pain during exercise show clear separation. Disease patients (red) cluster in higher ST depression zones (2-3mm+), indicating significant cardiac stress.

Hearts can hide problems at rest, but stress testing exposes cardiovascular vulnerabilities. The combination of exercise-induced symptoms plus ECG changes creates a powerful diagnostic signal.

*# FIGURE 3: ST Depression and Blood Pressure Interaction**# Demonstrates how exercise-induced changes relate to resting conditions*

```

fig3 <- ggplot(heart_data, aes(x = trestbps, y = oldpeak, color = disease)) +
  geom_point(alpha = 0.5, size = 2) +
  geom_density_2d(alpha = 0.3) +
  facet_wrap(~exang, labeller = labeller(exang = c("No Ex. Angina" = "No Exercise Angin
a",
                                                    "Ex. Angina" = "Exercise-Induced An
gina")))) +
  scale_color_manual(values = disease_colors, name = "Diagnosis") +
  labs(
    title = "Exercise Stress Response: ST Depression vs. Resting Blood Pressure",
    subtitle = "Exercise-induced angina combined with ST depression strongly indicates d
isease",
    x = "Resting Blood Pressure (mm Hg)",
    y = "ST Depression (mm)",
    caption = "Contour lines show density; higher ST depression suggests ischemia"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 10, color = "gray40"),
    strip.text = element_text(face = "bold", size = 11),
    legend.position = "bottom"
  )

print(fig3)

```

Exercise Stress Response: ST Depression vs. Resting Blood Pressure

Exercise-induced angina combined with ST depression strongly indicates disease



Contour lines show density; higher ST depression suggests ischemia

```
ggsave("figure3_stdepression_bp.png", fig3, width = 10, height = 6, dpi = 300)
```

Figure 4 - Cholesterol Distribution by Age Group and Disease Status

Figure 4 uses violin plots to show how cholesterol distributes across age groups. The width of each “violin” represents how common different cholesterol values are.

In younger patients (under 40), healthy and diseased individuals show similar, tight distributions. However, in middle age (40-60), disease patients show wider, more irregular distributions—suggesting disrupted or unpredictable cholesterol metabolism. By age 70+, distributions converge again, possibly due to survival bias and widespread medical treatment.

The variability and distribution pattern of cholesterol matters more than a single average value. Middle-aged disease patients show the most metabolic chaos.

*# FIGURE 4: Cholesterol Distribution by Age Group and Disease Status**# Uses violin plots to show distribution shapes*

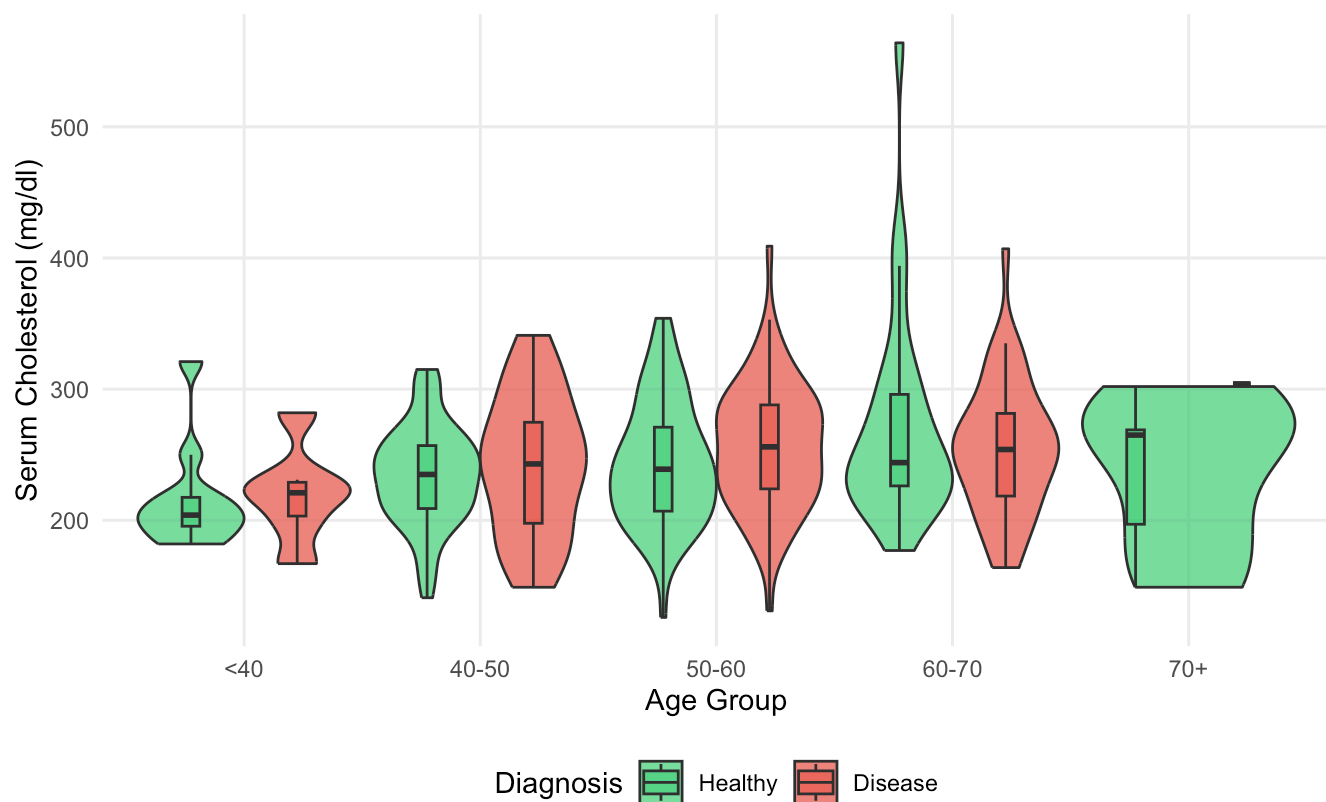
```
fig4 <- ggplot(heart_data, aes(x = age_group, y = chol, fill = disease)) +  
  geom_violin(alpha = 0.7, scale = "width", position = position_dodge(width = 0.9)) +  
  geom_boxplot(width = 0.15, position = position_dodge(width = 0.9),  
              alpha = 0.5, outlier.shape = NA) +  
  scale_fill_manual(values = disease_colors, name = "Diagnosis") +  
  labs(  
    title = "Cholesterol Levels Across Age Groups: Distribution Patterns",  
    subtitle = "Disease patients show wider cholesterol variability in middle age",  
    x = "Age Group",  
    y = "Serum Cholesterol (mg/dl)",  
    caption = "Violin width shows density; internal boxes show quartiles"  
  ) +  
  theme_minimal() +  
  theme(  
    plot.title = element_text(face = "bold", size = 14),  
    plot.subtitle = element_text(size = 10, color = "gray40"),  
    legend.position = "bottom",  
    panel.grid.minor = element_blank()  
  )  
  
print(fig4)
```

```
## Warning: Groups with fewer than two datapoints have been dropped.
```

```
## i Set `drop = FALSE` to consider such groups for position adjustment purposes.
```

Cholesterol Levels Across Age Groups: Distribution Patterns

Disease patients show wider cholesterol variability in middle age



Violin width shows density; internal boxes show quartiles

```
ggsave("figure4_cholesterol_violin.png", fig4, width = 10, height = 6, dpi = 300)
```

```
## Warning: Groups with fewer than two datapoints have been dropped.
## i Set `drop = FALSE` to consider such groups for position adjustment purposes.
```

Figure 5 - Major Vessels and Thalassemia Interaction Heatmap

Figure 5 is a risk stratification heatmap combining two factors: number of blocked major vessels (0-3) and thalassemia status (blood flow to heart muscle). Colors range from green (low risk) to red (high risk).

Patients with no blockages and normal blood flow show ~13-17% disease rates (green). But those with three blocked vessels AND reversible blood flow defects show 92% disease rates (deep red). The “reversible defect” means heart tissue is still alive but critically starved for blood.

These risk factors multiply rather than add. Having both blocked vessels and blood flow problems is exponentially more dangerous than either alone. This combination demands urgent intervention.

*# FIGURE 5: Major Vessels and Thalassemia Interaction Heatmap**# Shows the powerful predictive combination of these factors*

```

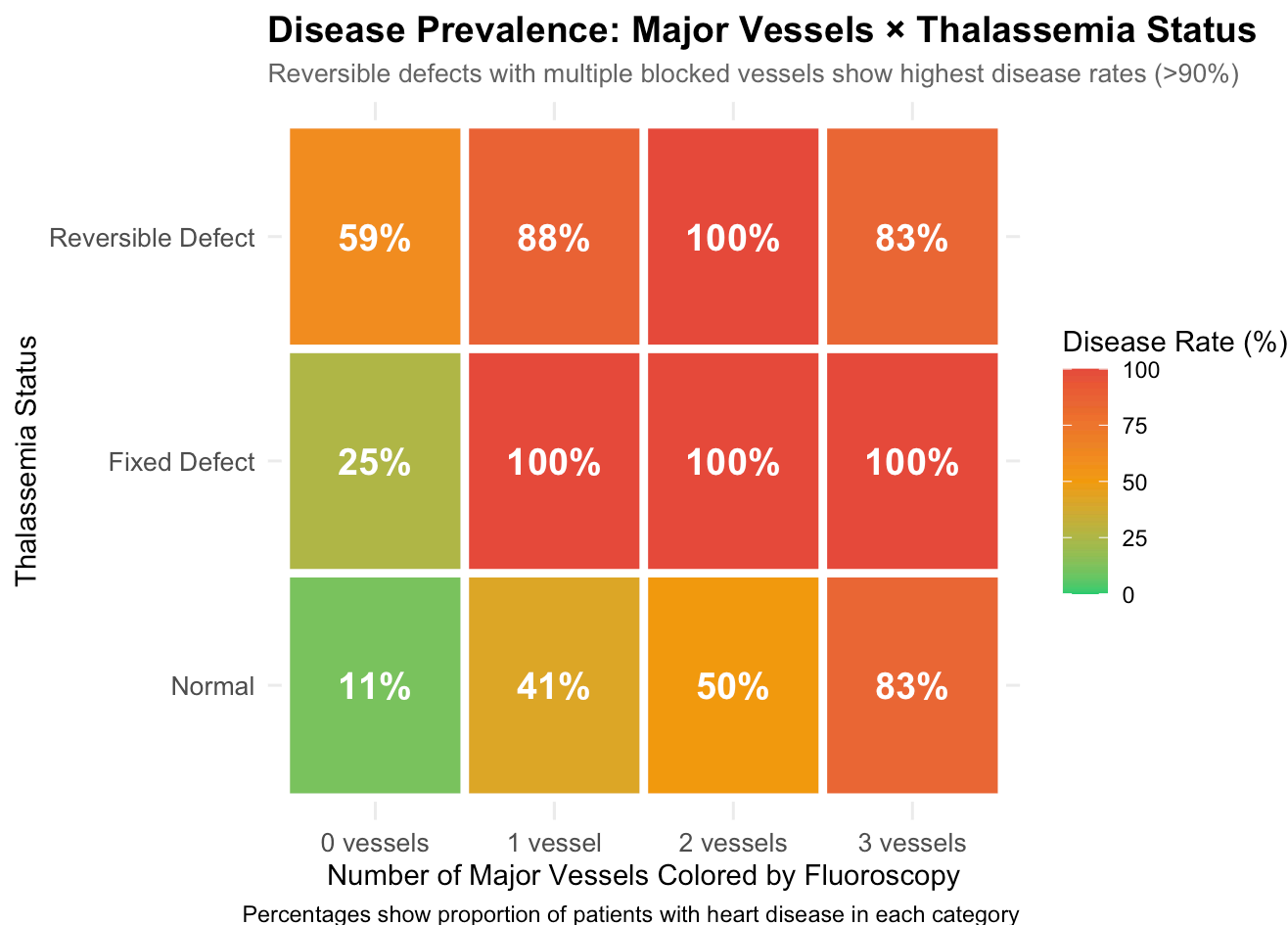
vessel_thal_summary <- heart_data %>%
  filter(!is.na(ca) & !is.na(thal)) %>%
  mutate(
    vessels = factor(ca, levels = 0:3, labels = c("0 vessels", "1 vessel", "2 vessels",
"3 vessels")),
    thal_type = factor(thal, levels = c(3, 6, 7),
      labels = c("Normal", "Fixed Defect", "Reversible Defect"))
  ) %>%
  group_by(vessels, thal_type, disease) %>%
  summarise(count = n(), .groups = "drop") %>%
  group_by(vessels, thal_type) %>%
  mutate(
    total = sum(count),
    disease_rate = count[disease == "Disease"] / total * 100
  ) %>%
  filter(disease == "Disease")

fig5 <- ggplot(vessel_thal_summary, aes(x = vessels, y = thal_type, fill = disease_rate)) +
  geom_tile(color = "white", size = 1.5) +
  geom_text(aes(label = sprintf("%.0f%%", disease_rate)),
    color = "white", fontface = "bold", size = 5) +
  scale_fill_gradient2(
    low = "#2ecc71", mid = "#f39c12", high = "#e74c3c",
    midpoint = 50, name = "Disease Rate (%)",
    limits = c(0, 100)
  ) +
  labs(
    title = "Disease Prevalence: Major Vessels x Thalassemia Status",
    subtitle = "Reversible defects with multiple blocked vessels show highest disease rates (>90%)",
    x = "Number of Major Vessels Colored by Fluoroscopy",
    y = "Thalassemia Status",
    caption = "Percentages show proportion of patients with heart disease in each category"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 10, color = "gray40"),
    axis.text = element_text(size = 10),
    legend.position = "right"
  )

```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
print(fig5)
```



```
ggsave("figure5_vessels_thal_heatmap.png", fig5, width = 10, height = 6, dpi = 300)
```

Figure 6 - Risk Factor Accumulation by Sex

Figure 6 tracks what happens when risk factors accumulate. We created a 0-5 risk score based on: exercise-induced chest pain, high blood sugar, abnormal ST slope, blocked vessels, and blood flow defects.

At 0-1 risk factors, most patients are healthy (green bars dominate). At 2 factors, disease rates increase but remain under 50%. At 3 factors, bars flip to majority red—disease becomes more likely than not. By 4-5 factors, disease rates exceed 80-90%.

Cardiovascular disease results from multiple failures across interconnected systems. This pattern holds for both sexes, indicating three risk factors represents a critical threshold requiring aggressive multi-targeted intervention.

FIGURE 6: Risk Factor Accumulation by Sex*# Demonstrates cumulative risk through multiple factors*

```

risk_data <- heart_data %>%
  mutate(
    risk_score = (as.numeric(exang) - 1) + # Exercise angina
                 (as.numeric(fbs) - 1) +   # High fasting blood sugar
                 (as.numeric(slope_type == "Flat" | slope_type == "Downsloping")) + # Ab
normal slope
                 (ca > 0) +                  # Any blocked vessels
                 (thal != 3)                 # Abnormal thalassemia
  ) %>%
  group_by(sex, risk_score, disease) %>%
  summarise(count = n(), .groups = "drop")

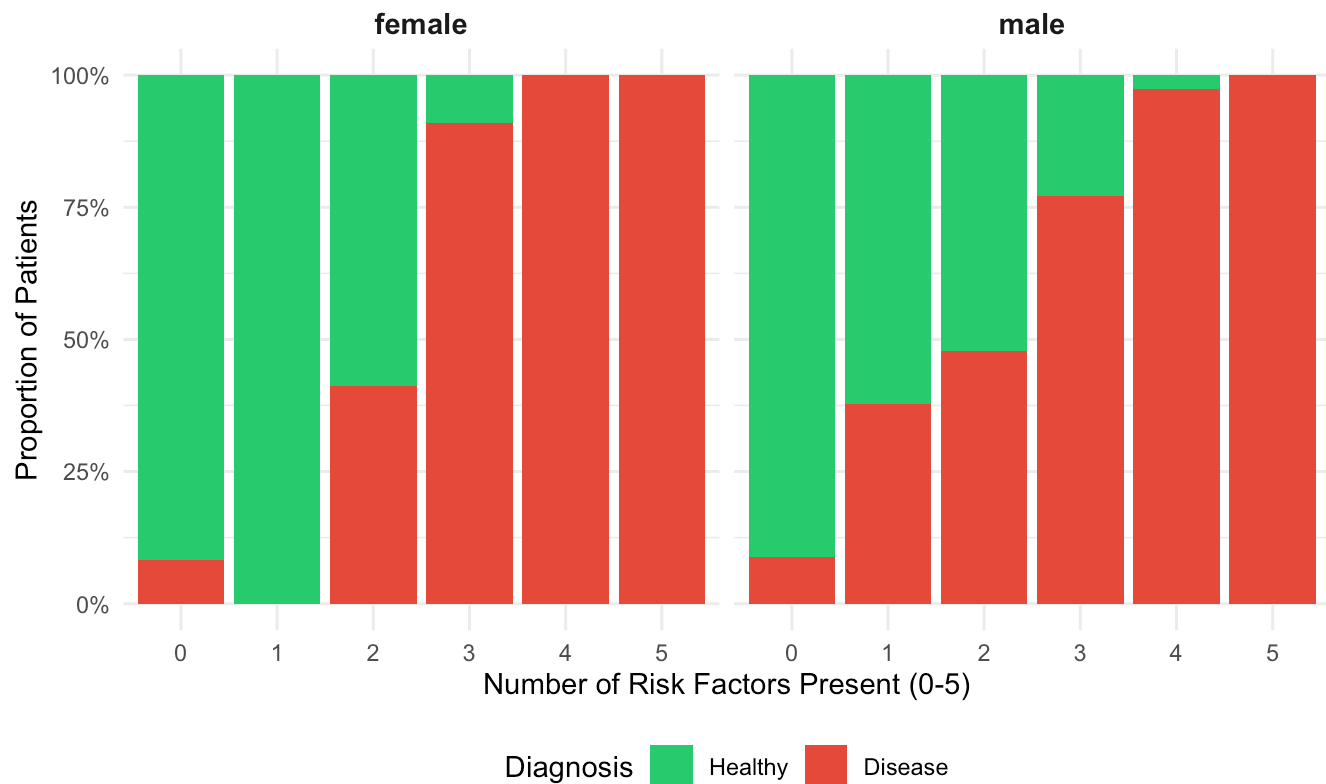
fig6 <- ggplot(risk_data, aes(x = factor(risk_score), y = count, fill = disease)) +
  geom_bar(stat = "identity", position = "fill") +
  facet_wrap(~sex, ncol = 2) +
  scale_fill_manual(values = disease_colors, name = "Diagnosis") +
  scale_y_continuous(labels = percent_format()) +
  labs(
    title = "Risk Factor Accumulation: The Compounding Effect",
    subtitle = "Disease prevalence increases dramatically with each additional risk fact
or",
    x = "Number of Risk Factors Present (0-5)",
    y = "Proportion of Patients",
    caption = "Risk factors: exercise angina, high FBS, abnormal slope, blocked vessels,
thalassemia defect"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 10, color = "gray40"),
    strip.text = element_text(face = "bold", size = 11),
    legend.position = "bottom"
  )

print(fig6)

```

Risk Factor Accumulation: The Compounding Effect

Disease prevalence increases dramatically with each additional risk factor



Risk factors: exercise angina, high FBS, abnormal slope, blocked vessels, thalassemia defect

```
ggsave("figure6_risk_accumulation.png", fig6, width = 10, height = 6, dpi = 300)
```

Figure 7 - ECG Results and Exercise Heart Rate

Figure 7 shows how resting ECG results (measured before any exercise) predict maximum heart rate capacity during stress testing.

Patients with normal resting ECGs achieve relatively good heart rates even when disease is present. Those with ST-T abnormalities show dramatic functional decline—disease patients average only 130-135 bpm. Left ventricular hypertrophy (LVH), where the heart muscle has thickened from chronic overwork, shows the worst outcomes—barely reaching 125-130 bpm maximum.

Abnormal resting ECGs don't just indicate disease presence; they predict functional impairment severity. Combined with disease, they signal a heart under serious strain.

FIGURE 7: ECG Results and Exercise Heart Rate

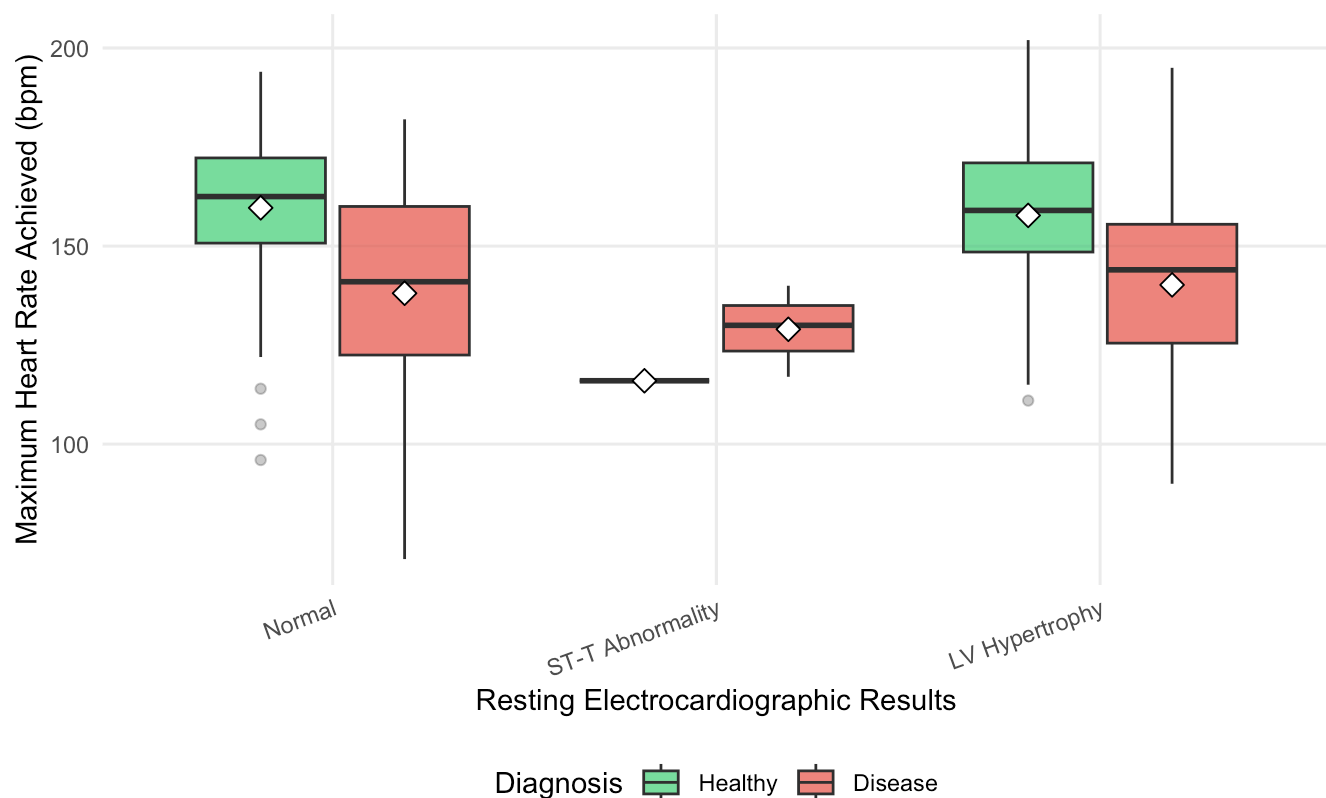
Shows relationship between resting ECG and exercise capacity

```
fig7 <- ggplot(heart_data, aes(x = restecg_type, y = thalach, fill = disease)) +
  geom_boxplot(alpha = 0.7, outlier.alpha = 0.3) +
  stat_summary(fun = mean, geom = "point", shape = 23, size = 3,
              position = position_dodge(width = 0.75),
              aes(group = disease), fill = "white") +
  scale_fill_manual(values = disease_colors, name = "Diagnosis") +
  labs(
    title = "Resting ECG Status vs. Exercise Heart Rate Capacity",
    subtitle = "Abnormal ECG correlates with reduced maximum heart rate in disease patients",
    x = "Resting Electrocardiographic Results",
    y = "Maximum Heart Rate Achieved (bpm)",
    caption = "Boxes show quartiles; diamond points show means"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 10, color = "gray40"),
    axis.text.x = element_text(angle = 20, hjust = 1),
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )

print(fig7)
```

Resting ECG Status vs. Exercise Heart Rate Capacity

Abnormal ECG correlates with reduced maximum heart rate in disease patients



Boxes show quartiles; diamond points show means

```
ggsave("figure7_ecg_heart_rate.png", fig7, width = 10, height = 6, dpi = 300)
```

Figure 8 - Comprehensive Age-Disease Profile Across Key Metrics

Figure 8 provides a comprehensive view by displaying four measurements simultaneously across age groups: maximum heart rate, cholesterol, resting blood pressure, and ST depression.

Maximum heart rate shows clear age-related decline, with disease patients consistently lower at all ages. Cholesterol distributions overlap substantially between healthy and diseased groups—it alone doesn't discriminate well. Blood pressure shows moderate elevation in disease patients. ST depression provides the starkest contrast: healthy patients cluster tightly near zero across all ages, while disease patients show wide distributions extending to 3-4mm.

No single measurement tells the complete story. ST depression and heart rate capacity are strong discriminators, while cholesterol is less definitive. Comprehensive multi-variable assessment provides the clearest diagnostic picture.

```
# FIGURE 8: Comprehensive Age-Disease Profile Across Key Metrics

# Ridge plot showing multiple variable distributions

library(ggribes)

age_profile <- heart_data %>%
  select(age_group, disease, thalach, chol, trestbps, oldpeak) %>%
  pivot_longer(cols = c(thalach, chol, trestbps, oldpeak),
               names_to = "metric", values_to = "value") %>%
  mutate(
    metric = factor(metric,
                    levels = c("thalach", "chol", "trestbps", "oldpeak"),
                    labels = c("Max Heart Rate (bpm)",
                              "Cholesterol (mg/dl)",
                              "Resting BP (mm Hg)",
                              "ST Depression (mm)"))
  )

fig8 <- ggplot(age_profile, aes(x = value, y = age_group, fill = disease)) +
  geom_density_ridges(alpha = 0.7, scale = 0.9) +
  facet_wrap(~metric, scales = "free_x", ncol = 2) +
  scale_fill_manual(values = disease_colors, name = "Diagnosis") +
  labs(
    title = "Age-Stratified Cardiovascular Profiles: A Comprehensive View",
    subtitle = "Disease patients show distinct patterns across multiple physiological measures",
    x = "Measurement Value",
    y = "Age Group",
    caption = "Density ridges show distribution shapes for each age group and diagnosis"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 10, color = "gray40"),
    strip.text = element_text(face = "bold", size = 10),
    legend.position = "bottom"
  )

print(fig8)
```

```
## Picking joint bandwidth of 7.59
```

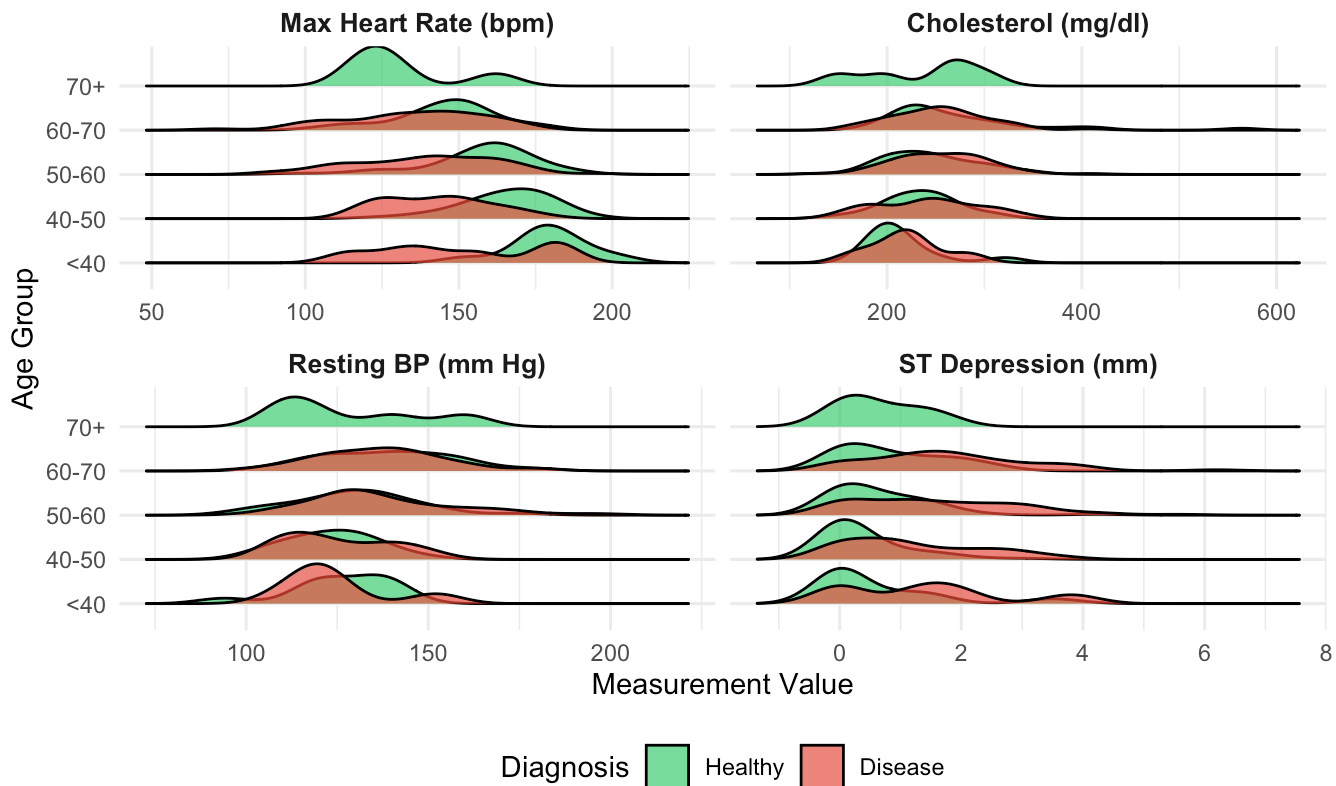
```
## Picking joint bandwidth of 19.8
```

```
## Picking joint bandwidth of 7.12
```

```
## Picking joint bandwidth of 0.452
```

Age-Stratified Cardiovascular Profiles: A Comprehensive View

Disease patients show distinct patterns across multiple physiological measures



Density ridges show distribution shapes for each age group and diagnosis

```
ggsave("figure8_age_profiles.png", fig8, width = 12, height = 8, dpi = 300)
```

```
## Picking joint bandwidth of 7.59
```

```
## Picking joint bandwidth of 19.8
```

```
## Picking joint bandwidth of 7.12
```

```
## Picking joint bandwidth of 0.452
```

Key Findings

This visual analysis reveals six critical insights about heart disease:

1. Maximum heart rate declines with age but is significantly lower in disease patients at all ages
2. Asymptomatic chest pain is paradoxically associated with higher disease prevalence
3. Exercise-induced ST depression is a strong disease indicator
4. Risk factors compound multiplicatively - patients with 3+ factors show >80% disease prevalence
5. The combination of blocked vessels and reversible thalassemia defects shows >90% disease rates

6. Sex differences exist in both disease prevalence and symptom presentation

Clinical Implications

For patients: Single blood tests aren't enough. Comprehensive evaluation including stress testing, ECG monitoring, and vessel imaging provides the full picture. The good news: most risk factors are modifiable through lifestyle changes and treatment.

For analysis: This demonstrates the power of multivariate visualization in uncovering complex health patterns. Each figure built on the previous, revealing layers of cardiovascular disease manifestation that single-variable analyses would miss.

Heart disease is complex but not mysterious. The patterns are clear when visualized properly we just need to know how to look.

```
# Disease prevalence by sex
prevalence <- heart_data %>%
  group_by(sex, disease) %>%
  summarise(count = n(), .groups = "drop") %>%
  group_by(sex) %>%
  mutate(percentage = count / sum(count) * 100)

print("Disease Prevalence by Sex:")
```

```
## [1] "Disease Prevalence by Sex:"
```

```
print(prevalence)
```

```
## # A tibble: 4 × 4
## # Groups:   sex [2]
##   sex    disease count percentage
##   <fct> <fct>   <int>      <dbl>
## 1 female Healthy    71        74.0
## 2 female Disease    25        26.0
## 3 male   Healthy    89        44.3
## 4 male   Disease   112        55.7
```

```
# Key correlations with disease
print("\nMean values by disease status:")
```

```
## [1] "\nMean values by disease status:"
```

```
summary_stats <- heart_data %>%  
  group_by(disease) %>%  
  summarise(  
    mean_age = mean(age),  
    mean_thalach = mean(thalach),  
    mean_oldpeak = mean(oldpeak),  
    mean_chol = mean(chol),  
    pct_with_exang = mean(as.numeric(exang) - 1) * 100  
  )  
print(summary_stats)
```

```
## # A tibble: 2 × 6  
##   disease mean_age mean_thalach mean_oldpeak mean_chol pct_with_exang  
##   <fct>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>  
## 1 Healthy    52.6         159.      0.599      243.       14.4  
## 2 Disease    56.8         139.      1.59       252.       54.0
```